Marek Chadim
marek.chadim@yale.edu

Problem 1. ***Inference with Stable Convergence.*** *Suppose that $Y_n \to_d \eta Z$ ($\mathcal{G}$-stably), where η is $\mathcal{G}$-measurable and is independent of $Z \sim N(0,1)$, and you have a consistent estimator $\hat{\eta}_n^2$ of the (possibly random) variance η^2: $\hat{\eta}_n^2 \to_p \eta^2$.*

(a) Show that $\begin{pmatrix} Y_n \\ \hat{\eta}_n^2 \end{pmatrix} \to_d \begin{pmatrix} \eta Z \\ \eta^2 \end{pmatrix}$ ($\mathcal{G}$-stably).

(b) Conclude that conditional on the event $\{\omega : \eta^2(\omega) > 0\}$, the t-statistic $Y_n/\sqrt{\hat{\eta}_n^2}$ is asymptotically $N(0,1)$ and that the confidence interval $Y_n \pm 1.96\sqrt{\hat{\eta}_n^2}$ covers 0 with probability tending to 0.95.

Solution.

(a) $\mathcal{G}$-stable convergence of Y_n implies joint convergence with every $\mathcal{G}$-measurable random variable. Since η^2 is $\mathcal{G}$-measurable, $(Y_n, \eta^2) \to_d (\eta Z, \eta^2)$ ($\mathcal{G}$-stably).

It remains to replace η^2 with $\hat{\eta}_n^2$. Fix $E \in \mathcal{G}$, continuity points (y, v), and $\epsilon > 0$. The inclusions

$$\{Y_n \le y,\, \eta^2 \le v - \epsilon\} \cap E \;\subseteq\; \{Y_n \le y,\, \hat{\eta}_n^2 \le v\} \cap E \;\cup\; \{|\hat{\eta}_n^2 - \eta^2| > \epsilon\},$$
$$\{Y_n \le y,\, \hat{\eta}_n^2 \le v\} \cap E \;\subseteq\; \{Y_n \le y,\, \eta^2 \le v + \epsilon\} \cap E \;\cup\; \{|\hat{\eta}_n^2 - \eta^2| > \epsilon\}$$

together with $\mathbb{P}(|\hat{\eta}_n^2 - \eta^2| > \epsilon) \to 0$ yield

$$\begin{aligned} \lim_n \mathbb{P}(\{Y_n \le y,\, \eta^2 \le v - \epsilon\} \cap E) &\le \liminf_n \mathbb{P}(\{Y_n \le y,\, \hat{\eta}_n^2 \le v\} \cap E) \\ &\le \limsup_n \mathbb{P}(\{Y_n \le y,\, \hat{\eta}_n^2 \le v\} \cap E) \\ &\le \lim_n \mathbb{P}(\{Y_n \le y,\, \eta^2 \le v + \epsilon\} \cap E). \end{aligned}$$

The outer limits exist by $\mathcal{G}$-stable convergence of (Y_n, η^2). Sending $\epsilon \to 0$ and using continuity at v, the bounds coincide, so $(Y_n, \hat{\eta}_n^2) \to_d (\eta Z, \eta^2)$ $\mathcal{G}$-stably.

(b) The event $\{\eta^2 > 0\} \in \mathcal{G}$ since η is $\mathcal{G}$-measurable. By part (a), the joint convergence holds conditional on any $E \in \mathcal{G}$ with $\mathbb{P}(E) > 0$; in particular on $\{\eta^2 > 0\}$. Define $g(x, v) = x/\sqrt{v}$ for $v > 0$, which is continuous on the support of $(\eta Z, \eta^2)$ conditional on $\eta^2 > 0$. By the continuous mapping theorem,

$$\frac{Y_n}{\sqrt{\hat{\eta}_n^2}} \;\to_d\; \frac{\eta Z}{\sqrt{\eta^2}} = \frac{\eta Z}{\eta} = Z \sim N(0,1)$$

conditional on $\{\eta^2 > 0\}$, using $\eta \ge 0$. Since $Y_n \pm 1.96\sqrt{\hat{\eta}_n^2}$ covers 0 iff $|Y_n/\sqrt{\hat{\eta}_n^2}| \le 1.96$,

$$\mathbb{P}\big(0 \in Y_n \pm 1.96\sqrt{\hat{\eta}_n^2} \;\big|\; \eta^2 > 0\big) \;\to\; \mathbb{P}(|Z| \le 1.96) = 0.95.$$

□

Problem 2. ***Bartik Instruments.*** *Consider $Y_i = \beta X_i + u_i$ with $X_i = \sum_{k=1}^K W_{ik} G_{ik}$, where $G_{ik} = G_k + \tilde{G}_{ik}$. The instrument is $Z_i = \sum_{k=1}^K V_{ik} G_k$, where V_{ik} are exposures from an earlier time period. Conditioning on $G_1, \ldots, G_K$, treat $(Y_i, \tilde{G}_i, V_i, W_i)_{i=1}^n$ as IID draws, with the exogeneity condition $\mathbb{E}[V_i u_i \mid G_1, \ldots, G_K] = 0$.*

(a) Consider the IV estimator $\hat{\beta} = \frac{\sum_{i=1}^n Z_i Y_i}{\sum_{i=1}^n Z_i X_i}$. Derive the distribution of $\sqrt{n}(\hat{\beta} - \beta)$ as $n \to \infty$. State any additional assumptions used.

(b) Using the result from (a), propose and formally justify a 95% confidence interval for β.

Solution.

(a) Substituting $Y_i = \beta X_i + u_i$:

$$\sqrt{n}(\hat{\beta} - \beta) = \frac{n^{-1/2} \sum_{i=1}^n Z_i u_i}{n^{-1} \sum_{i=1}^n Z_i X_i}.$$

Let $\mathcal{G} = \sigma(G_1, \ldots, G_K)$ and $G = (G_1, \ldots, G_K)'$, so $Z_i = V_i' G$.

Conditional on $\mathcal{G}$, the random variables $\{Z_i u_i\}_{i=1}^n$ are IID with mean $\mathbb{E}[Z_i u_i \mid G] = \sum_{k=1}^K G_k \, \mathbb{E}[V_{ik} u_i \mid G] = G' \mathbb{E}[V_i u_i \mid G] = 0$ by the exogeneity condition.

The following additional assumptions are imposed: $\mathbb{E}[Z_i^2 u_i^2] < \infty$ (finite variance for the CLT). $\mathbb{E}[Z_i X_i \mid G] \neq 0$ a.s. (relevance condition). $\mathbb{E}[Z_i^2], \mathbb{E}[X_i^2] < \infty$ (for the LLN).

Conditional on G, $\{Z_i u_i\}$ are IID mean-zero with variance $\sigma^2(G) := \mathbb{E}[Z_i^2 u_i^2 \mid G]$. By the conditional CLT, $\frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i u_i \;\to_d\; \sigma(G) \cdot Z$ ($\mathcal{G}$-stably), where $Z \sim N(0,1)$ is independent of G.

By the conditional LLN, $n^{-1} \sum Z_i X_i \to_p \mathbb{E}[Z_i X_i \mid G] =: \mu(G) \neq 0$ a.s. By a Slutsky extension for $\mathcal{G}$-stable convergence (as in Problem 1(a)):

$$\sqrt{n}(\hat{\beta} - \beta) \;\to_d\; \frac{\sigma(G)}{\mu(G)} Z \quad (\mathcal{G}\text{-stably}), \qquad \eta^2 = \frac{\mathbb{E}[Z_i^2 u_i^2 \mid G]}{(\mathbb{E}[Z_i X_i \mid G])^2}.$$

Conditional on G, the limit is $N(0, \eta^2)$. Unconditionally, it is a mixture of normals.

(b) Define $\hat{u}_i = Y_i - \hat{\beta} X_i$ and the variance estimator $\hat{V}_n = \frac{n^{-1} \sum_{i=1}^n Z_i^2 \hat{u}_i^2}{\left(n^{-1} \sum_{i=1}^n Z_i X_i\right)^2}$. Since $\hat{\beta} \to_p \beta$, the conditional LLN gives $n^{-1} \sum Z_i^2 \hat{u}_i^2 \to_p \mathbb{E}[Z_i^2 u_i^2 \mid G]$, so $\hat{V}_n \to_p \eta^2$. By Problem 1(b) with $Y_n = \sqrt{n}(\hat{\beta} - \beta)$ and $\hat{\eta}_n^2 = \hat{V}_n$, conditional on $\{\eta^2 > 0\}$:

$$\frac{\sqrt{n}(\hat{\beta} - \beta)}{\sqrt{\hat{V}_n}} \;\to_d\; N(0,1).$$

The 95% confidence interval is

$$\hat{\beta} \;\pm\; 1.96 \cdot \frac{\left(n^{-1} \sum_{i=1}^n Z_i^2 \hat{u}_i^2\right)^{1/2}}{|n^{-1} \sum_{i=1}^n Z_i X_i| \cdot \sqrt{n}},$$

with coverage tending to 0.95 conditional on $\{\eta^2 > 0\}$ by Problem 1(b).

□

Problem 3. ***LLN for Dyadic Data.*** *Suppose you observe a $n \times n$ array of dyadic data $(Y_{t_1,t_2})_{1\le t_1,t_2\le n}$. The data are from a stationary random field Y indexed by $\mathbb{Z}^2$. You may assume $|Y_{t_1,t_2}| \le C$ a.s. for all t_1, t_2.*

(a) Derive a summability condition on the strong mixing coefficients $\alpha(c; m, n)$ of Y under which a LLN holds:

$$\frac{1}{n^2}\sum_{t_1,t_2=1}^{n} Y_{t_1,t_2} \;\to_p\; \mathbb{E}[Y_{t_1,t_2}].$$

For the mixing coefficients, use $d(t,s) = \max(|t_1 - s_1|, |t_2 - s_2|)$.

(b) How does the summability condition change if $d(t,s) = \min(|t_1 - s_1|, |t_2 - s_2|)$?

Solution.

(a) WLOG assume $\mathbb{E}[Y_t] = 0$. Let $\bar{Y}_n = n^{-2}\sum_t Y_t$. By Chebyshev's inequality, it suffices to show $\mathrm{Var}(\bar{Y}_n) \to 0$. Since $|Y_t| \le C$, the covariance inequality for α-mixing gives $|\mathrm{Cov}(Y_t, Y_s)| \le 4C^2\alpha(d(t,s);\, 1, 1)$. Substituting $h = t - s$ and bounding $(n - |h_1|)(n - |h_2|) \le n^2$:

$$\mathrm{Var}(\bar{Y}_n) \le \frac{4C^2}{n^2}\sum_{c=0}^{n-1} N_{\max}(c)\,\alpha(c;\, 1, 1),$$

where $N_{\max}(c) = |\{h \in \mathbb{Z}^2 : \max(|h_1|, |h_2|) = c\}|$. A direct count gives $N_{\max}(0) = 1$ and $N_{\max}(c) = (2c+1)^2 - (2c-1)^2 = 8c$ for $c \ge 1$. Therefore

$$\mathrm{Var}(\bar{Y}_n) \le \frac{4C^2}{n^2}\left[\alpha(0;\, 1, 1) + 8\sum_{c=1}^{n-1} c\,\alpha(c;\, 1, 1)\right].$$

This tends to zero provided the summability condition $\sum_{c=1}^{\infty} c\,\alpha(c;\, 1, 1) < \infty$ holds.

(b) With $d(t,s) = \min(|t_1 - s_1|, |t_2 - s_2|)$, the covariance bound becomes $|\mathrm{Cov}(Y_t, Y_s)| \le 4C^2\,\alpha(\min(|h_1|, |h_2|);\, 1, 1)$. The shell volume changes: $\min(|h_1|, |h_2|) = c$ requires both coordinates $\ge c$ in absolute value with at least one equal to c, giving

$$N_{\min}(c) = 4(n - c) + 4(n - c - 1) = 8(n - c) - 4 \le 8n.$$

Therefore

$$\mathrm{Var}(\bar{Y}_n) \le \frac{4C^2}{n^2}\sum_{c=0}^{n-1} N_{\min}(c)\,\alpha(c;\, 1, 1) \le \frac{32C^2}{n}\sum_{c=0}^{n-1}\alpha(c;\, 1, 1).$$

This tends to zero provided $\sum_{c=0}^{\infty}\alpha(c;\, 1, 1) < \infty$. The factor of c disappears because $N_{\min}(c) = O(n)$ rather than $O(c)$: with $d_{\min}$, many more lattice points lie at each distance level. In fact, $\sum\alpha(c;\, 1, 1) < \infty$ is sufficient but stronger than necessary. By Cesàro's theorem, $\alpha(c;\, 1, 1) \to 0$ implies $\frac{1}{n}\sum_{c=0}^{n-1}\alpha(c;\, 1, 1) \to 0$, so *$\alpha$-mixing alone suffices*. The natural sufficient summability condition thus weakens from $\sum c\,\alpha$ in part (a) to $\sum\alpha$ here, reflecting the larger shell volumes under $d_{\min}$.

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